

Table 4 - Hyperparameter search spaces used in Optuna TPE optimization

Model	Trials	Hyperparameter	Type	Range
RF	150	n_estimators	integer	[200, 1000]
RF	150	max_depth	integer	[3, 20]
RF	150	min_samples_leaf	integer	[1, 20]
RF	150	min_samples_split	integer	[2, 20]
RF	150	max_features	categorical	{sqrt, log2, 0.3, 0.5, 0.7}
RF	150	bootstrap	categorical	{True, False}
RF	150	max_samples (if bootstrap)	real	[0.5, 1.0]
XGBoost	150	n_estimators	integer	[200, 1000]
XGBoost	150	max_depth	integer	[2, 10]
XGBoost	150	learning_rate	real (log)	[1e-3, 0.3]
XGBoost	150	subsample	real	[0.5, 1.0]
XGBoost	150	colsample_bytree	real	[0.5, 1.0]
XGBoost	150	reg_alpha	real (log)	[1e-8, 100]
XGBoost	150	reg_lambda	real (log)	[1e-8, 100]
XGBoost	150	min_child_weight	integer	[1, 10]
CatBoost	150	iterations	integer	[200, 1000]
CatBoost	150	depth	integer	[3, 8]
CatBoost	150	learning_rate	real (log)	[1e-3, 0.3]
CatBoost	150	l2_leaf_reg	real (log)	[1e-2, 10]
CatBoost	150	bagging_temperature	real	[0.0, 1.0]
CatBoost	150	border_count	integer	[32, 255]
LightGBM	150	n_estimators	integer	[200, 1000]
LightGBM	150	num_leaves	integer	[15, 127]
LightGBM	150	max_depth	integer	[3, 12]
LightGBM	150	learning_rate	real (log)	[1e-3, 0.3]
LightGBM	150	min_child_samples	integer	[5, 50]
LightGBM	150	subsample	real	[0.5, 1.0]
LightGBM	150	colsample_bytree	real	[0.5, 1.0]
LightGBM	150	reg_alpha	real (log)	[1e-8, 10]
LightGBM	150	reg_lambda	real (log)	[1e-8, 10]